

P4B Solar Mini-Grid: Socioeconomic Impact Analysis

CS 6150 – Computing for Good
Fall 2024

Background

Context: In early 2024, Renewvia published the first-ever peer-reviewed, comprehensive socioeconomic impact analysis for mini-grids in rural Africa. The data is open-source, and the statistical tools used in the paper to draw conclusions are all very general. Little attention was paid to subgroup analyses or correlations in results. There is potential for follow-up analyses to ask and answer questions related to the impact of adding solar mini-grids on communities.

Goal: Analyze the data from the analysis to understand the impact of Renewvia's work on local communities.

Project Chronology

- **Sep. 4** – Team formed based on interested in socioeconomic impact analysis.
- **Sep. 16** – Team discussion of roles and task delegation within team based on individual skills and preferences.
- **Sep. 19** – Our team had a meeting with Dr. Selby from our partner organization, Renewvia to discuss the study and to complete handover of the study data for our team to analyze.
- **Oct. 6** – Team completed exploratory data analysis for education and healthcare variables.

Project Chronology

- **Oct. 30** – Over multiple meetings with Dr. Selby, we decided to continue onward with education as our primary subgroup focus and settled on a letter deliverable that could be published to a journal.
- **Nov. 10** – We previously sent an outline for the paper to Dr. Selby and received suggestions for improvement. Our team is currently working on writing the paper to be finalized and submitted for publishing by early December.

Implementation of Course Principles

Respect for Stakeholder's Time and Feedback & Collaborative Development:

Our team maintains consistent communication with our partner through emails and meetings to relay progress and gain feedback to integrate into our project.

Simple, Intuitive, and Minimal UI:

The website requires little to no training for the user to understand how to navigate the user interface. Our project pertains mainly to showcasing data and analyses conducted by our team. The website contains common elements like a navigation bar, making it more intuitive for users to navigate. There is little interaction required from the user.

Immediate and Continuous Use for Users:

The data, charts, and related elements are all available on the project website. The user can extrapolate the data and conclusion from our team's analyses immediately. In addition, there is information included on how our analyses were conducted, allowing for users to apply our methods to other groups for further research.

Introduction:

- Talk about Renewvia and the general work they do.
- Discuss data collection and how it was collected
- Discuss our focus on solar mini-grids for improving education in Kenya and Nigeria

Methods:

- Data cleaning and preprocessing
- Describing how we converted the post survey data into binary and ordinal variables.
- Describe the principle component analysis to get the concise dataset of 26 variables.
- Model selection: We use the school_performance_change as the dependent variable. We use the logistic regression(ridge), logistic regression (Lasso), random forest, gradient boosting and SVM 5 model approaches to help us to filter the key components that make significant impact to the dependent variables. Through the 5 above model, we filter out 8 variables that make significant impact to the dependent variables.
- statistical test: Then based on the different variable types, we did statistic tests to check if these variables show significant difference within different dependent variables group. We exclude the variables that failed in statistic tests and focus on the variables passed the tests
- selected feature analysis: We analyze the power and direction of the filtered variables, explore how these variables set up connection with school performance change in reality, and how they were impacted by mini-grid projects. Then we summarize our findings to analyze how mini-grid impacted the education of the local communities directly and indirectly.

Results:

- Discussion of key indicators from the analysis, such as home exterior lighting, recent business activity, and changes in household headcount, illustrate factors within the household environment that appear to significantly influence educational performance.
- Economic stability, home security, and household stability show to all contribute meaningfully to children's educational outcomes.
- Show how this demonstrates that the mini-grid project impacts educational performance not only by providing a stable energy supply and reliable lighting access that enables extended study hours, but by enhancing the overall living environment, fostering economic stability, and creating job opportunities to improve the broader socioeconomic environment.

Discussion and Potential Future Work:

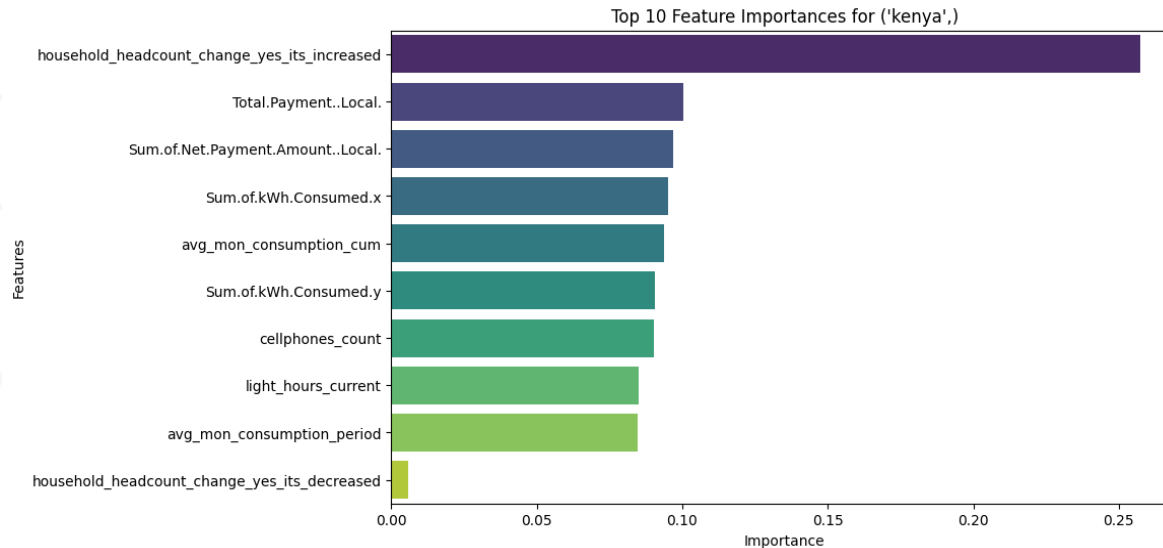
- How this affects Renewvia's work.
- Potential future areas of analysis that could continue research

Letter Outline

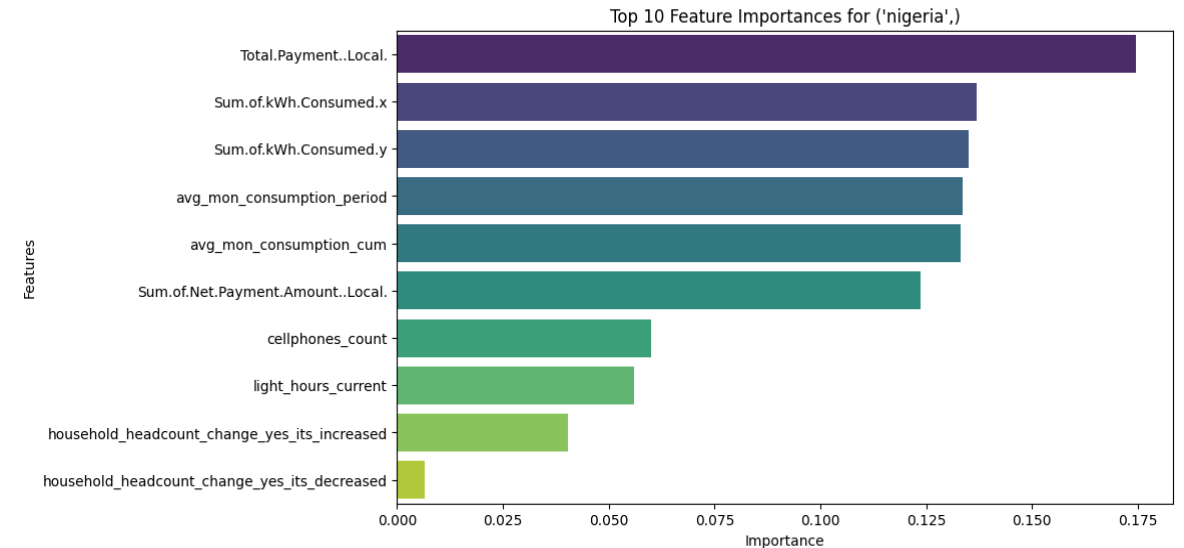
This outline was presented to Dr. Selby as a sample rough draft for the proposed letter for ACS Electronic Letters. Each bullet point represents a paragraph in the finished paper. The final paper should be about 3-6 pages long including references to previous letters on the socioeconomic impact of mini solar-grids.

Analysis: Education

- Kenya



- Nigeria



Kenya and Nigeria were studied to evaluate education improvements through solar mini-grids. The first step in this analysis is feature selection. To select features, we used a technique called Recursive Feature Elimination (RFE), which helps systematically identify the most important features from a larger set by repeatedly training the model. The charts above represent the top features for Kenya and Nigeria. This feature importance illustrates how much each factor contributes to predicting school performance, which reflects educational improvement.

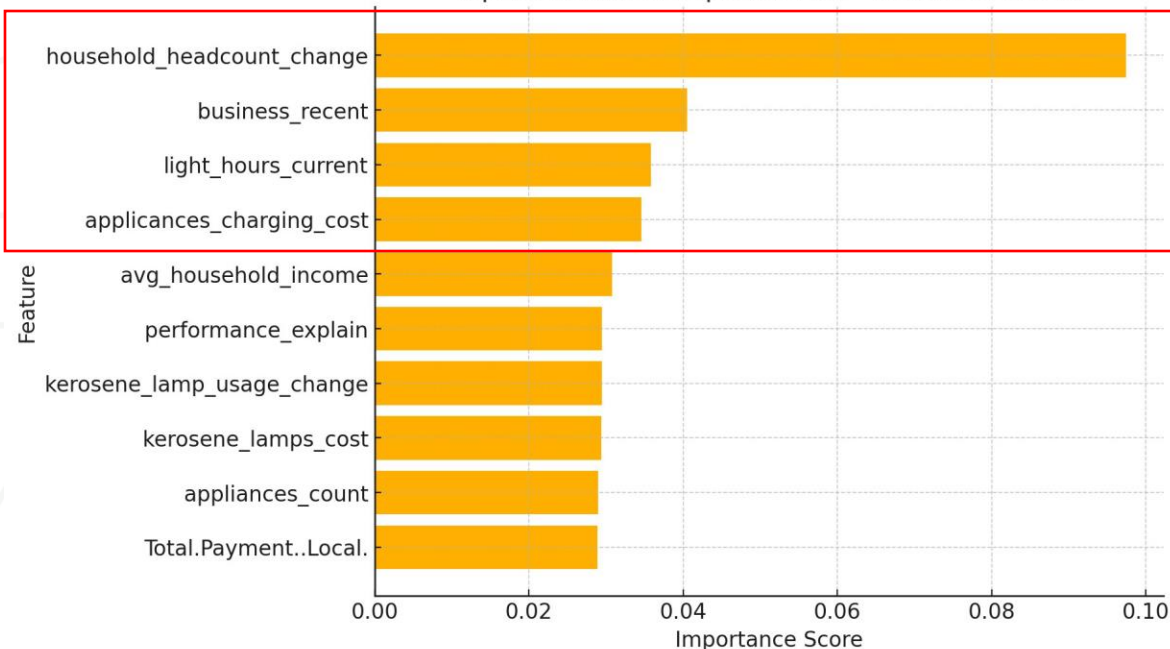
Analysis: Education-Data Cleaning

A	B	C	D	E	F	G	H	I	J	K	L	M
occupation_change	household_income_change	household_headcount_change	girls_schooling_change	boys_schooling_change	school_performance_change	school_perform	business_recent	business_from_minigrid	business_use_minigrid	power_sources	power_sources_usage	power_sources_primary
N	no_it_is_the_same	no	N	N	N		no	yes	no	minigrid_electricity_renewvia;		5 M
N	no_it_is_the_same	no	N	N	N		no	yes	no	charcoal;		2 M
Y	no_it_is_the_same	no	I	N	N		yes	yes	yes	solar_home_kit;not_available ;		2 not_available
N	no_it_is_the_same	no	N	N	N		no	yes	no	minigrid_electricity_renewvia;		5 M
N	no_it_is_the_same	no	N	N	N		no	no	no	minigrid_electricity_renewvia;		8 M
Y	no_it_is_the_same	no	N	N	N		no	yes	no	minigrid_electricity_renewvia;		12 M
Y	yes_it_has_increased	I	N	N	N		no	yes	no	minigrid_electricity_renewvia;		8 M
Y	no_it_is_the_same	no	I	I	N		yes	yes	no	solar_home_kit;minigrid_electricity_renewvia;		8 M
Y	no_it_is_the_same	no	N	N	N		no	yes	yes	minigrid_electricity_renewvia;		2 M
N	no_it_is_the_same	no	N	N	N		no	yes	no	minigrid_electricity_renewvia;		8 M
Y	no_it_is_the_same	no	N	N	N		yes	yes	yes	minigrid_electricity_renewvia;		8 M
Y	yes_it_has_increased	no	I	N	N		no		no	solar_home_kit;		5 M
N	no_it_is_the_same	no	N	N	N		no	yes	no	minigrid_electricity_renewvia;		8 M
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N	no_it_is_the_same	no	N	N	yes_its_gotten_better	because_we_h	no	no	no	solar_home_kit;		8 solar_home_kit
Y	no_it_is_the_same	no	I	I	yes_its_gotten_better	not_available	yes	yes	no	solar_home_kit;		8 M
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N	no_it_is_the_same	no	N	N	N		no	yes	no	minigrid_electricity_renewvia;		8 M
Y	yes_it_has_increased	I	N	N	yes_its_gotten_better	increased_study	yes	yes	yes	minigrid_electricity_renewvia;		8 M
N	no_it_is_the_same	no	N	N	N		yes	yes	no	kerosene_lamp;		2 M
N	no_it_is_the_same	I	N	N	yes_its_gotten_better	because_they_f	no	yes	no	solar_home_kit;		5 M
Y	yes_it_has_increased	no	N	N	N		no	yes	no	minigrid_electricity_renewvia;		12 M
N	no_it_is_the_same	no	N	N	N		no	yes	no	minigrid_electricity_renewvia;		2 M
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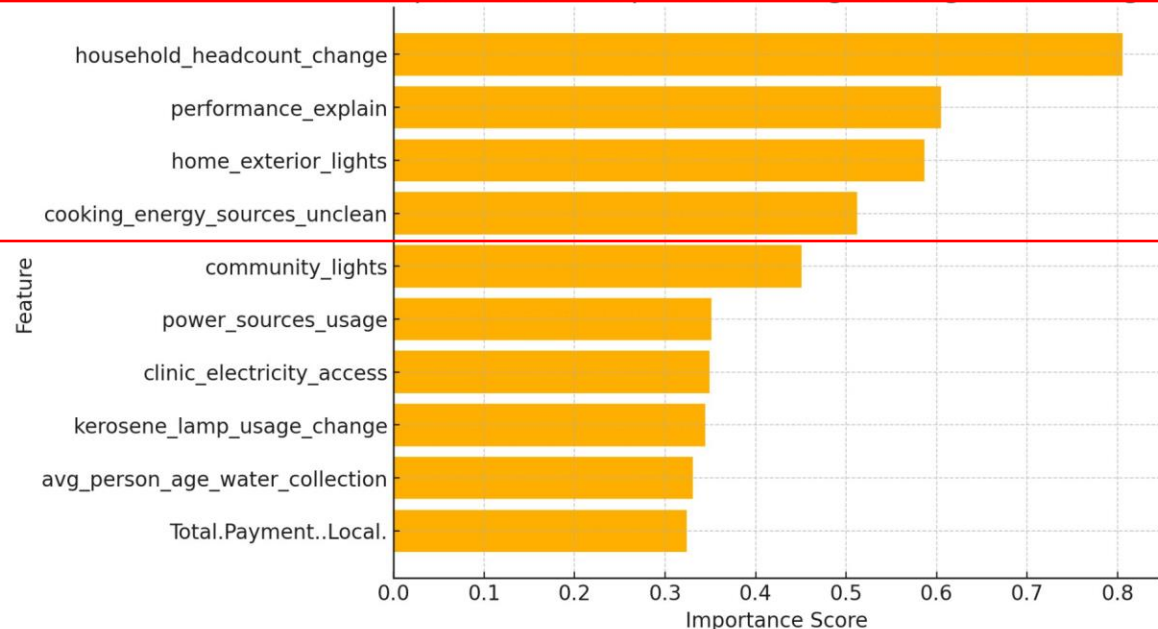
- We exclude all text variable but the school performance change reason and make the following change to variables:
For columns with 2 choices (Such as yes, no) , converting into binary variable For columns with 3 choices (Such as Yes,increased/No same/Yes, decreased), converting into ordinal variable (like I/N/D).
For numerical variable (such as energy usage) keep it as its original value
For column with number scope (such as 100-200, 200-500,500-1000), choose the ceiling value.
- Then we do a principal component analysis to combine some variables measured similar aspects. And we get 26 components finally.

Analysis: Education-Key Features

Top 10 Feature Importances - Random Forest



Top 10 Feature Importances - Logistic Regression (Ridge)



The Machine learning models helped us to filter 8 key variables that show strong relationship with the dependent variable: school performance changing, they are:

Total Energy Payment (Local),

Total Energy Consumption (kWh),

Appliance Count,

cooking_energy_source_unclean ,

home_exterior_lights,

business_recent,

household_headcount_change,

performance_explain

Analysis: Education-Summary

	Coef.	Std.Err.	Z-value	P> Z
Intercept	-4.0512	0.2340	-17.3151	0.0000
Total_Energy_Payment_Local	0.0000	0.0000	-3.1161	0.0018
Total_Energy_Consumption_kWh	0.0009	0.0007	1.3476	0.1778
Appliance_Count	0.0340	0.0168	2.0216	0.0432
home_exterior_lights	1.3734	0.1900	7.2303	0.0000
business_recent	1.0648	0.1781	5.9783	0.0000
cooking_energy_source_unclean	1.3679	0.2242	6.1006	0.0000
community_lights	0.5356	0.1747	3.0655	0.0022
performance_explain	1.3495	0.1645	8.2059	0.0000
household_headcount_change	-2.1347	0.2005	-10.6448	0.0000

The result indicates that Mini-grid project impact the education directly and indirectly in local community.

For example, the

Those Significant predictors such as Home Exterior Lights and light hours current show how Mini-grid provided the stable energy supply and more access to light.

Other predictors, like Business Recent, Cooking Energy Source (Unclean), and Household Headcount Change, highlight factors in the household environment that influence school performance. Mini-grid help to improve the economic stability (business activity), home security (exterior lights), and household stability (headcount), which all play substantial roles in supporting children's education outcomes.